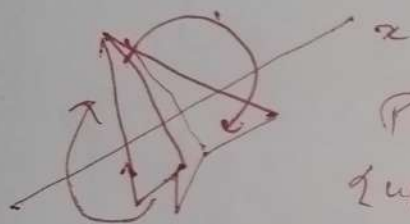
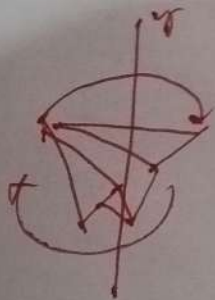


Culer Angles

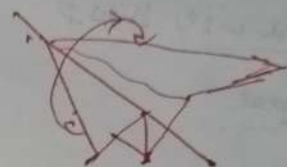
- Pitch { Rotation about x-axis }
- Yaw { Rotation about y-axis }
- Roll { Rotation about z-axis }



Pitch
{ up-down movement }



Yaw
{ Right - left movement }



Roll
{ tilting }

Transformations

$$\text{front} = R_y(\text{yaw}) * (1, 0, 0)$$

$$\text{Right} = \text{normalize}(\text{front} \times \text{world up})$$

$$\text{Pitch} \text{ around the right axis}$$

$$c_y = \cos(\text{yaw})$$

$$s_y = \sin(\text{yaw})$$

$$c_p = \cos(\text{pitch})$$

$$s_p = \sin(\text{pitch})$$

$$\text{Yaw} = \begin{bmatrix} \cos y & 0 & -\sin y \\ 0 & 1 & 0 \\ \sin y & 0 & \cos y \end{bmatrix}$$

$$\text{front} = \text{Yaw} \cdot (1, 0, 0)$$

$$= \begin{bmatrix} \cos y & 0 & -\sin y \\ 0 & 1 & 0 \\ \sin y & 0 & \cos y \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \cos y \\ 0 \\ \sin y \end{bmatrix} = (\cos y, 0, \sin y)$$

Now

$$\text{Right} = \text{front} \times \text{World up}$$

$$= (\cos y, 0, \sin y) \times (0, 1, 0)$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos y & 0 & \sin y \\ 0 & 1 & 0 \end{vmatrix}$$

$$= (-\sin y) \hat{i} - 0 \hat{j} + \cos y \hat{k}$$

$$\Rightarrow (-\sin y, 0, \cos y)$$

Now, we rotate along this arbitrary axis Right, we get matrix

$$m = \begin{bmatrix} c_p + s_y^2(1-c_p) & -c_y s_p & -s_y c_y(1-c_p) \\ c_y s_p & c_p & s_y s_p \\ -s_y c_y(1-c_p) & -s_y s_p & c_p + c_y^2(1-c_p) \end{bmatrix}$$

now, find vector

$$v = m \times \text{front}$$

We get.

$$x = c_y c_p$$

$$y = s_p$$

$$z = c_y c_p$$

$$x = \cos(\text{yaw}) \cos(\text{pitch})$$

$$y = \sin(\text{pitch})$$

$$z = \sin(\text{yaw}) \times \cos(\text{pitch})$$